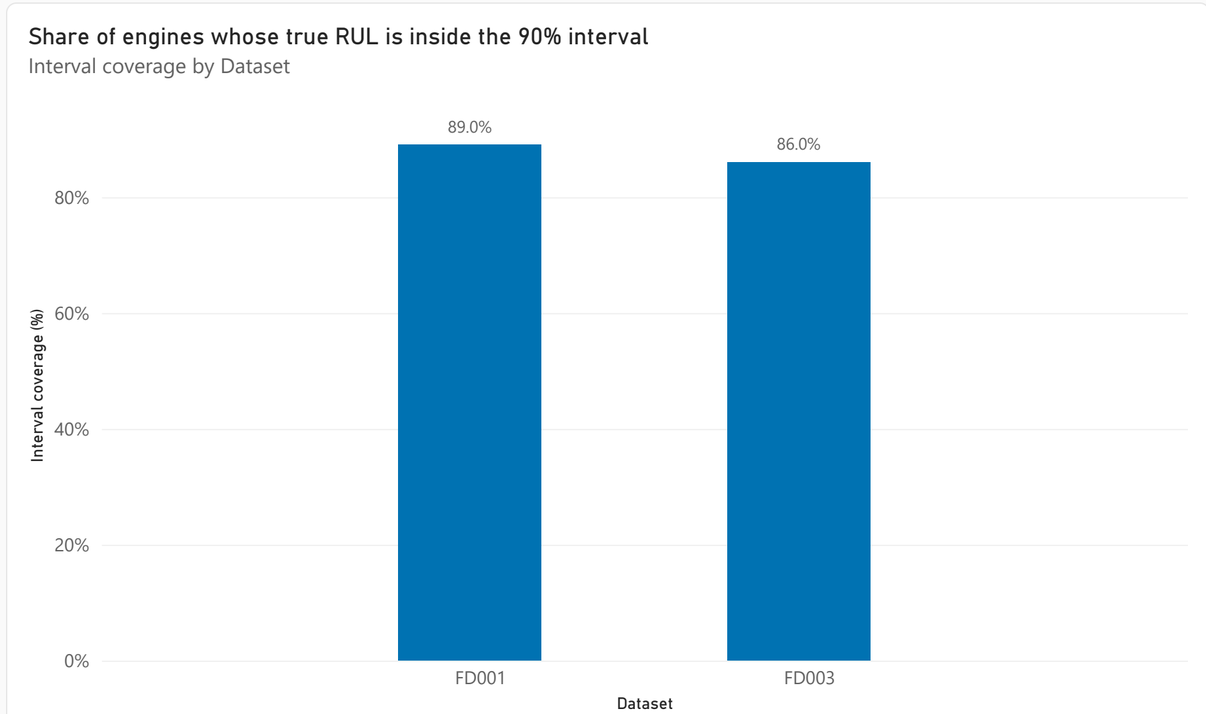
Which engines are closest to failure, and how sure is the model?

fleet_summary

Dataset	Engines	Flagged	Interval coverage
FD001	100	23	89.0%
FD003	100	20	86.0%

Flagged: the lower 90% bound on remaining useful life (RUL) is 20 cycles or less at the engine's last cycle.
Nominal interval: 90%. FD001 coverage is post hoc (rule chosen after FD001 test results were seen); FD003 is the pre-registered confirmation.



Which engines should the crews service in the next 7 days?

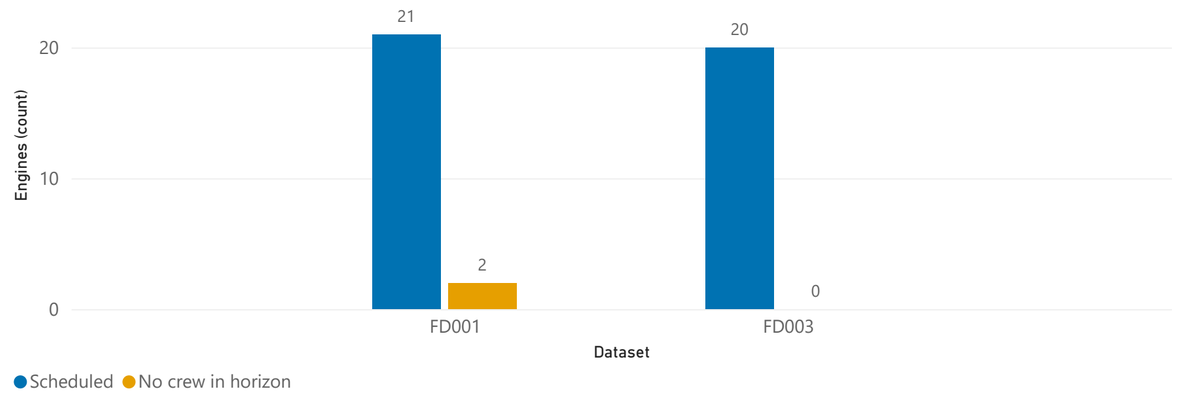
plan_summary

Dataset	Engines flagged	Scheduled	No crew in horizon
FD001	23	21	2
FD003	20	20	0

Schedule from OR-Tools CP-SAT under the crew limit; engines with no crew slot inside the 7-day horizon are listed as 'flagged, no crew in horizon'.

Flagged engines: scheduled vs no crew slot in 7 days

Scheduled and No crew in horizon by Dataset



Flagged engines and their service day (end-of-data plan)

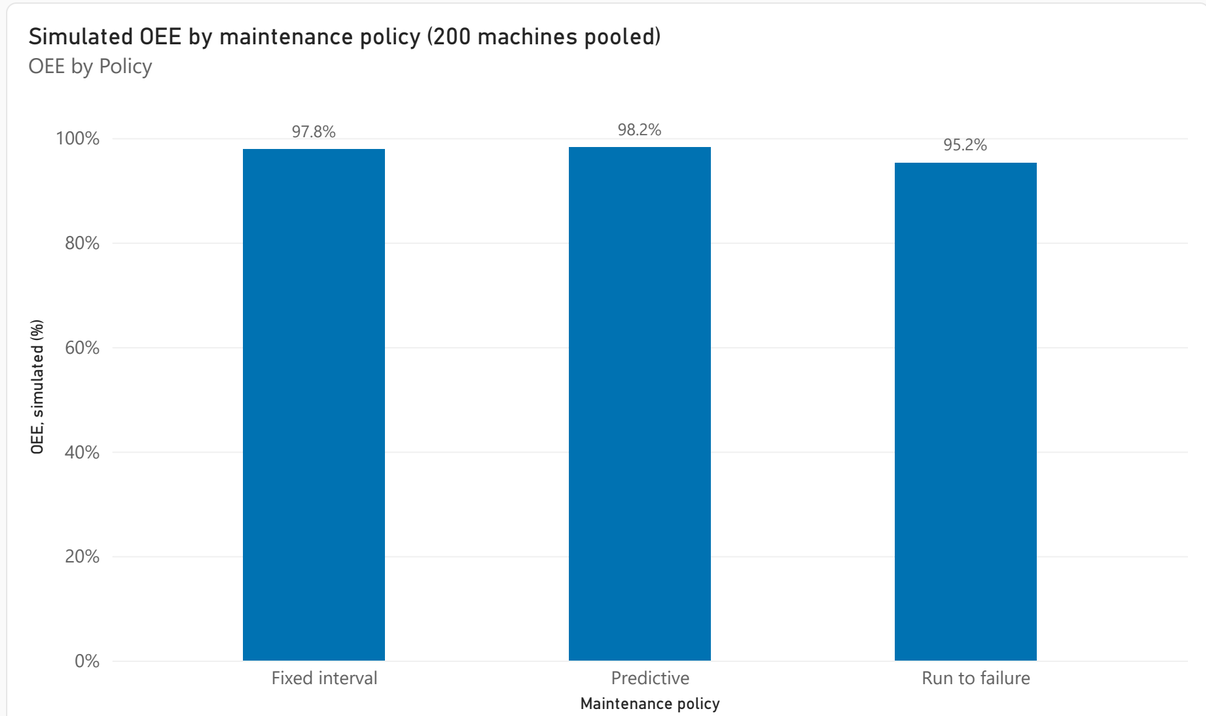
Dataset	Engine	Risk	Lower RUL bound (cycles)	Service day	Status
FD001	20	high	9.53	4	serviced in time
FD001	24	medium	12.50	3	serviced in time
FD001	31	critical	4.92	1	serviced in time
FD001	34	critical	0.00	0	serviced in time
FD001	35	critical	2.44	1	serviced in time
FD001	36	medium	13.66	5	serviced in time
FD001	37	medium	16.33	6	serviced in time
FD001	41	medium	11.49	5	serviced in time
FD001	42	critical	1.65	0	serviced in time
FD001	49	medium	10.37	3	serviced in time
FD001	56	high	5.97	2	serviced in time
FD001	61	medium	16.06	6	serviced in time
FD001	64	medium	17.85		flagged, no crew in horizon
FD001	66	medium	12.27	2	serviced in time
FD001	68	high	6.05	2	serviced in time

Which maintenance policy gives the highest OEE? (Simulated plant)

oee_policies

Policy	OEE	Unplanned failures	Planned services
Fixed interval	97.8%	21	179
Predictive	98.2%	8	192
Run to failure	95.2%	200	0

Simulated: the plant, its hours and units are a simulation driven by the NASA engine data; they are not measurements from a real plant.
OEE = availability x performance x quality, each a ratio of sums over the machines.



Can we trust the model's forecasts this shift?

monitoring_summary

Scenario	Dataset	Runs	Flagged runs	Flag rate	C1 result
phase7_v1	FD001	484	15	3.1%	
phase7_v1	FD003	484	45	9.3%	
phase7b_v2_staggered	FD002	667	6	0.9%	PASS
phase7b_v2_staggered	FD004	667	93	13.9%	FAIL
phase7b_v2_synchronized	FD002	554	0	0.0%	PASS
phase7b_v2_synchronized	FD004	554	108	19.5%	FAIL

Phase 7 drift false-alarm rate

11.6%

Phase 7 drift false-alarm rate

Phase 7 target (5%)

not met

Phase 7 target result

C1 (pre-registered for Phase 7b): clean-run flag rate at most 5%. It passes on FD002 and FAILS on FD004.
Phase 7's drift false-alarm target was not met. Both are reported, not re-tuned.

Share of clean runs flagged for retraining, by dataset and scenario

Flag rate by Dataset and Scenario

